

Sample Quiz 3

DATA 412/612

1. Create the following matrix:

	[,1]	[,2]	[,3]	[,4]	[,5]	[,6]	[,7]	[,8]	[,9]	[,10]
[1,]	990	980	970	960	950	940	930	920	910	900
[2,]	890	880	870	860	850	840	830	820	810	800
[3,]	790	780	770	760	750	740	730	720	710	700
[4,]	690	680	670	660	650	640	630	620	610	600
[5,]	590	580	570	560	550	540	530	520	510	500
[6,]	490	480	470	460	450	440	430	420	410	400
[7,]	390	380	370	360	350	340	330	320	310	300
[8,]	290	280	270	260	250	240	230	220	210	200
[9,]	190	180	170	160	150	140	130	120	110	100
[10,]	90	80	70	60	50	40	30	20	10	0

2. Find the sum of the entries of the even columns of the above matrix.

3. Suppose you have the following data frame:

```
df <- data.frame(
  Name = c("Johnny", "Alisson", "Andy", "Sarah", "Alex"),
  Age = c(25, 30, 28, 22, 32),
  Annual_Income = c(50000, 60000, 55000, 48000, 70000),
  Account_Balance = c(1700, 2000, 1800, 900, 2500.90)
)
```

- What is the age average of people who get more than \$51k in a year.
 - What are their names?
4. Given a data frame `df` with only positive values, write a function that check if the number of columns in it is less than equal to 1 or it has negative values at all. If yes, it returns -1 . If not, the function compares the geometric means (using the pipe operator) of the first and the second columns and produces 1 if that is correct, otherwise 0. Try your function on these inputs `data.frame(a=c(9,1,8))`, `data.frame(a=c(9,9,8), b=c(9,7,6))`, and `data.frame(a=c(9,1,8), b=c(9,17,6))`.